Avnit Singh

Mohali, Punjab | avnitsingh58@gmail.com | 8766576639 | LinkedIn | GitHub | Portfolio

Professional Summary

Full-Stack Software Engineer (Next.js, TypeScript, Node.js, PostgreSQL) shipping production features and high-volume BigQuery telemetry pipelines for US-based SaaS platforms. Skilled in low-latency REST APIs, event-driven backends (Redis, BullMQ), and AI-driven systems (LLMs, AI agents) alongside scalable React architectures.

Experience

Software Engineer Intern *Mar 2026 – Present*
Code Garage Tech *Mohali, Punjab*

- **Creative Launch Engine:** Engineered ad orchestration workflows in Next.js 15 & TypeScript, automating multi-variant launches across Meta/TikTok Ads and cutting campaign setup time by 40%.
- **Telemetry & Data Resilience:** Architected a BigQuery ingestion pipeline processing 25,000+ daily events, pairing `sendBeacon` with a GCS Dead Letter Queue (DLQ) and cron replay to achieve 99.9% data durability.
- **AI Compliance Guardrails:** Integrated Google Gemini to build an automated pre-flight policy verification pipeline, checking scripts against advertising guidelines to slash manual review turnaround by 75%.
- **Performance & Multi-Tenant RBAC:** Reduced dashboard P95 response times from 2.5s to <200ms via a PostgreSQL caching layer over BigQuery, while eliminating UI race conditions (AbortController) and enforcing Clerk RBAC.

MERN Stack Developer Trainee *Jul 2025 – Dec 2025*
CertED Technologies *Roorkee, Uttarakhand*

- Built a full-stack blogging platform from scratch to master core MERN principles, architecting modular REST APIs with Express.js, request validation, and clean MVC separation.
- Designed database models for posts and categories with parent-child relationships to render nested reply threads and manage real-time interactions like post likes.
- Secured workflows with dual-token JWT authentication and HTTP-only cookies, combining custom authorization middleware with a responsive React UI and image upload handling.

Skills

- **Frontend & Languages:** TypeScript, JavaScript, SQL, React.js, Next.js, Redux, Tailwind CSS, HTML5, CSS3
- **Backend & Architecture:** Node.js, Express.js, REST APIs, Redis, BullMQ, JWT, Clerk, Rate Limiting
- **Databases & ORM:** PostgreSQL, Prisma ORM, Google BigQuery, MongoDB (Mongoose ODM)
- **AI & Data Engineering:** LLM Integration (Gemini, OpenAI), AI Agents
- **Cloud, Tools & DevOps:** Docker, Git, GitHub, Vercel, Sentry, Postman

Projects

CanaryGuard – Canary Verification & Rollback Platform GitHub | Live Demo
Node.js, TypeScript, BullMQ, Redis, Express, PostgreSQL, Prisma, Docker

- Engineered a dynamic reverse proxy and BullMQ/Redis worker pipeline routing live HTTP traffic with <2ms overhead, scaling releases across 4 stages (5% → 25% → 50% → 100%) while limiting defect blast radius to <5% of users.
- Built an SLA evaluation engine using a 60-second in-memory sliding window for P99 latency and 5xx error spikes, executing automated rollbacks in <15 seconds and restoring 100% traffic to baseline.
- Developed an autonomous Node.js AI diagnostic agent that ingests stderr logs, metric snapshots, and git commit diffs, diagnosing defects with <10s detection latency and 85%+ root-cause confidence.

Writely – Full-Stack Content Publishing & Discussion Platform GitHub | Live Demo
Next.js, Node.js, Express, PostgreSQL, Sequelize, JWT, Zod

- Designed a normalized PostgreSQL schema with Sequelize ORM; cut discussion thread query latency by 45% using a single-fetch adjacency hash map that parses nested replies in $O(n)$ time without N+1 queries.
- Implemented dual-token JWT auth with silent token refresh, HTTP-only secure cookies, and centralized Axios interceptors for session persistence and XSS/CSRF mitigation.
- Built a responsive Next.js frontend with schema-driven Zod forms and Multer image uploads; enforced composite unique constraints to guarantee zero duplicate likes under concurrent requests.

Education

Bachelor of Technology in Computer Science and Engineering *Sept 2022 – June 2026*
Uttarakhand Technical University (VMSBUTU)